

## SWINOUJSCIE, Poland

WMO: 122000

Lat: 53.9233N

Lon: 14.2422E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.3      | -7.5 | -13.4                         | 1.2 | -9.7 | -10.4 | 1.5 | -6.4 | 13.8                  | 2.7  | 12.2 | 1.5  | 2.5                   | 110  | 0.545 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.3                    | 27.1            | 19.3 | 25.0 | 18.9 | 23.4 | 18.3 | 21.0                | 25.0 | 20.0 | 23.4 | 19.1 | 22.3 | 3.0                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 19.6                            | 14.3 | 22.8 | 18.7 | 13.5 | 21.9 | 17.9 | 12.8 | 21.0 | 60.5          | 25.1 | 57.2 | 23.4 | 54.4 | 22.3 | 26.9              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 11.3          | 9.3  | 7.7 | DB                         | -12.5 | 32.4               | 4.2 | 2.6                                                | -15.5 | 34.2       | -17.9 | 35.7       | -20.3 | 37.2       | -23.3 | 39.0 | (4) |
| (5)               |      |     | WB                         | -12.8 | 22.8               | 4.1 | 1.3                                                | -15.8 | 23.7       | -18.2 | 24.5       | -20.5 | 25.2       | -23.5 | 26.2 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.2         | 0.4   | 1.3   | 3.7   | 7.9   | 12.2  | 15.8  | 18.2  | 18.3  | 14.7  | 9.9   | 5.2   | 1.9   |  |  |
|                                                                        | DBStd       | 7.24        | 4.32  | 4.04  | 3.64  | 3.68  | 3.30  | 2.84  | 2.67  | 2.64  | 2.76  | 3.18  | 3.22  | 4.12  |  |  |
|                                                                        | HDD10.0     | 1274        | 297   | 243   | 196   | 83    | 13    | 0     | 0     | 0     | 1     | 41    | 147   | 253   |  |  |
|                                                                        | HDD18.3     | 3424        | 556   | 476   | 453   | 312   | 193   | 86    | 35    | 33    | 113   | 261   | 396   | 511   |  |  |
|                                                                        | CDD10.0     | 974         | 0     | 0     | 2     | 21    | 82    | 174   | 254   | 258   | 142   | 39    | 2     | 0     |  |  |
|                                                                        | CDD18.3     | 82          | 0     | 0     | 0     | 1     | 3     | 11    | 31    | 33    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 406         | 0     | 0     | 0     | 7     | 29    | 77    | 142   | 135   | 16    | 0     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 90          | 0     | 0     | 0     | 1     | 6     | 22    | 30    | 30    | 1     | 0     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.2         | 3.3   | 3.3   | 3.5   | 3.6   | 3.4   | 3.2   | 3.2   | 2.9   | 3.1   | 3.1   | 3.0   | 3.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 586         | 42    | 34    | 37    | 32    | 52    | 62    | 71    | 63    | 54    | 45    | 46    | 46    |  |  |
|                                                                        | PrecMax     | 728         | 79    | 70    | 80    | 91    | 106   | 121   | 251   | 165   | 108   | 91    | 101   | 96    |  |  |
|                                                                        | PrecMin     | 471         | 0     | 5     | 11    | 3     | 14    | 18    | 2     | 22    | 24    | 18    | 3     | 10    |  |  |
|                                                                        | PrecStd     | 79          | 22    | 14    | 18    | 18    | 24    | 26    | 48    | 28    | 24    | 19    | 20    | 20    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.6  | 11.6  | 16.0  | 24.1  | 26.9  | 29.9  | 30.2  | 30.3  | 25.6  | 19.9  | 14.0  | 11.5  |  |  |
|                                                                        |             | MCWB        | 8.6   | 9.1   | 10.3  | 14.3  | 17.4  | 20.0  | 20.5  | 21.3  | 17.8  | 15.4  | 12.4  | 10.4  |  |  |
|                                                                        | 2%          | DB          | 8.7   | 9.6   | 12.4  | 19.0  | 22.4  | 25.3  | 26.6  | 26.4  | 22.8  | 17.3  | 12.0  | 9.8   |  |  |
|                                                                        |             | MCWB        | 7.5   | 7.8   | 9.0   | 12.4  | 15.6  | 18.4  | 19.9  | 19.5  | 17.2  | 14.4  | 10.6  | 8.8   |  |  |
|                                                                        | 5%          | DB          | 7.2   | 8.4   | 10.4  | 15.7  | 19.6  | 22.5  | 24.6  | 24.4  | 20.9  | 15.7  | 10.6  | 8.4   |  |  |
|                                                                        |             | MCWB        | 6.1   | 6.6   | 7.8   | 11.1  | 14.8  | 17.3  | 19.5  | 19.0  | 16.4  | 13.5  | 9.4   | 7.2   |  |  |
|                                                                        | 10%         | DB          | 5.9   | 7.0   | 8.8   | 13.4  | 17.5  | 20.7  | 22.9  | 22.9  | 19.2  | 14.4  | 9.5   | 7.0   |  |  |
|                                                                        |             | MCWB        | 5.0   | 5.6   | 6.7   | 9.9   | 13.8  | 16.5  | 18.7  | 18.4  | 15.7  | 12.7  | 8.5   | 6.1   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 9.3   | 9.7   | 11.3  | 15.5  | 18.6  | 20.8  | 22.3  | 22.6  | 19.6  | 16.5  | 12.8  | 10.7  |  |  |
|                                                                        |             | MCDB        | 10.2  | 10.9  | 14.2  | 22.5  | 25.0  | 26.7  | 26.6  | 27.3  | 22.7  | 18.6  | 13.7  | 11.2  |  |  |
|                                                                        | 2%          | WB          | 7.6   | 8.3   | 9.5   | 13.3  | 16.7  | 19.2  | 21.1  | 20.9  | 18.1  | 14.9  | 10.8  | 8.9   |  |  |
|                                                                        |             | MCDB        | 8.5   | 9.4   | 11.9  | 17.5  | 20.8  | 23.7  | 25.0  | 24.7  | 21.0  | 16.5  | 11.7  | 9.6   |  |  |
|                                                                        | 5%          | WB          | 6.3   | 6.9   | 8.1   | 11.6  | 15.5  | 18.1  | 20.2  | 19.9  | 17.1  | 13.8  | 9.6   | 7.4   |  |  |
|                                                                        |             | MCDB        | 7.1   | 8.1   | 10.1  | 14.9  | 18.8  | 21.6  | 23.6  | 23.0  | 19.7  | 15.3  | 10.5  | 8.2   |  |  |
|                                                                        | 10%         | WB          | 5.0   | 5.5   | 6.9   | 10.3  | 14.3  | 17.1  | 19.2  | 19.1  | 16.1  | 12.9  | 8.5   | 6.1   |  |  |
|                                                                        |             | MCDB        | 5.9   | 6.8   | 8.6   | 13.1  | 17.1  | 20.2  | 22.3  | 22.0  | 18.8  | 14.2  | 9.4   | 6.9   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 3.8   | 4.5   | 5.8   | 6.9   | 7.5   | 7.6   | 7.3   | 7.7   | 6.8   | 5.6   | 3.9   | 3.5   |  |  |
|                                                                        |             | MCDBR       | 4.4   | 6.0   | 8.8   | 11.3  | 11.7  | 11.5  | 10.4  | 10.5  | 9.5   | 7.2   | 4.8   | 4.3   |  |  |
|                                                                        | 5% WB       | MCWBR       | 4.0   | 4.5   | 5.7   | 6.0   | 6.0   | 5.7   | 5.0   | 5.2   | 4.9   | 4.7   | 3.8   | 3.9   |  |  |
|                                                                        |             | MCWBR       | 4.3   | 5.5   | 8.1   | 10.0  | 10.4  | 10.0  | 8.8   | 8.7   | 8.3   | 6.7   | 4.6   | 4.3   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.319 | 0.343 | 0.373 | 0.408 | 0.406 | 0.401 | 0.423 | 0.420 | 0.385 | 0.365 | 0.343 | 0.314 |  |  |
|                                                                        |             | taud        | 2.329 | 2.326 | 2.275 | 2.245 | 2.310 | 2.351 | 2.302 | 2.318 | 2.392 | 2.398 | 2.366 | 2.321 |  |  |
|                                                                        | Edn at Noon |             | 635   | 743   | 798   | 817   | 842   | 849   | 820   | 798   | 781   | 709   | 599   | 560   |  |  |
|                                                                        |             | Edh at Noon | 58    | 80    | 104   | 122   | 120   | 117   | 120   | 112   | 92    | 74    | 57    | 49    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 0.53  | 1.18  | 2.42  | 4.17  | 5.28  | 5.60  | 5.17  | 4.31  | 3.12  | 1.61  | 0.66  | 0.37  |  |  |
|                                                                        | RadStd      |             | 0.06  | 0.18  | 0.25  | 0.53  | 0.61  | 0.42  | 0.64  | 0.36  | 0.33  | 0.23  | 0.09  | 0.04  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | +0.38 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (44 neighbors) |  | +0.40 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -149    | N/A     | N/A     |

Nomenclature: See separate page